Lab 1 – Requirements Engineering & UML Use-Case Modelling

#09 – Faculty Research Grant & Publication Tracker

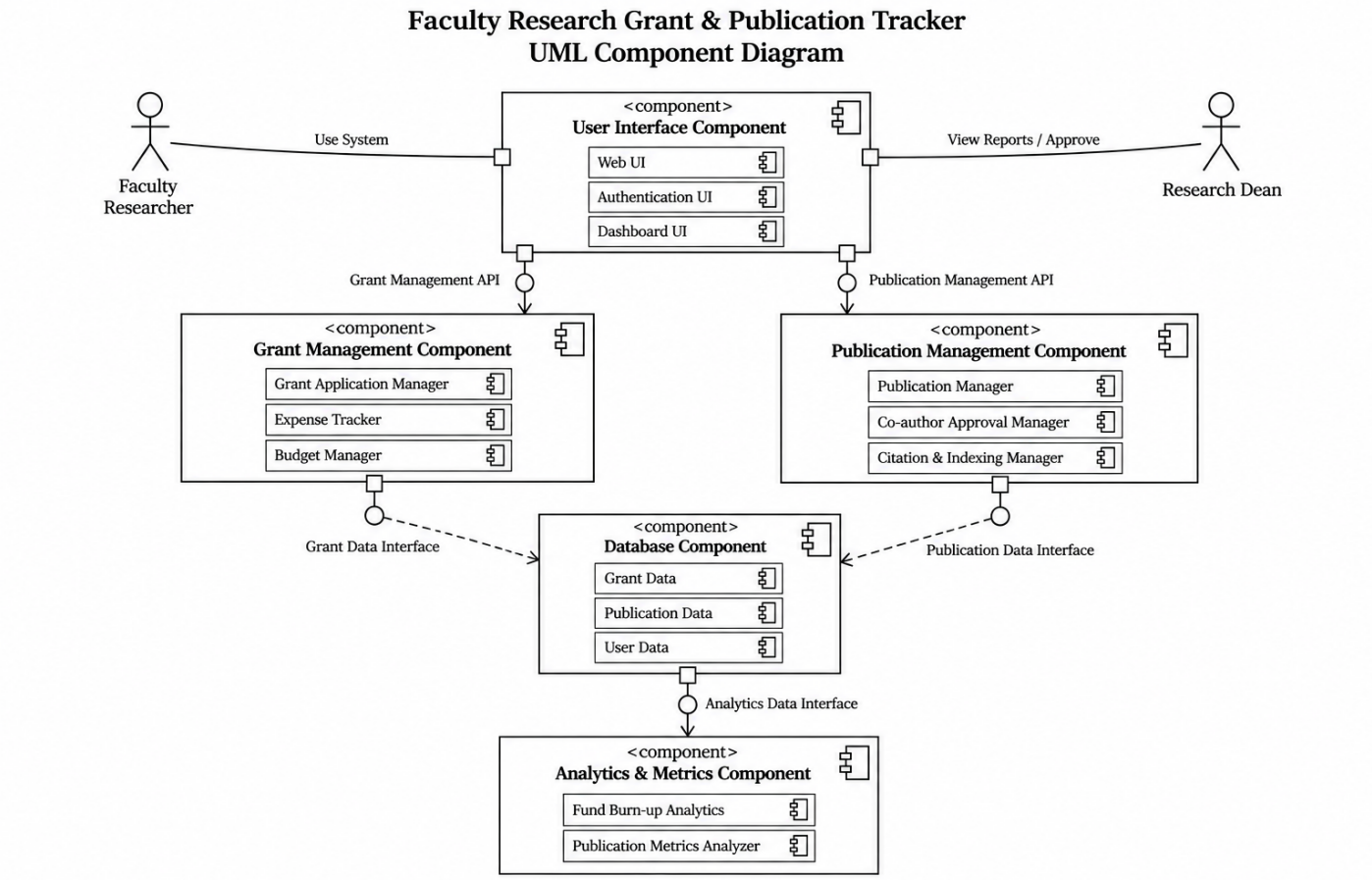
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall track grant fund allocations, allow faculty to log equipment procurement expenses, and calculate the remaining grant budget.	High	Pass: Remaining balance updates accurately and expenses exceeding the grant cap are blocked. Fail: Overdraft is allowed without admin override.	Prevents grant overspending and provides accurate financial tracking.	Sample requirement from problem statement.
FR-002	Functional	The system shall allow faculty researchers to record and manage publication details, including title, authors, venue, and publication status.	High	Pass: Publication details are saved and can be retrieved or updated. Fail: Required publication information cannot be saved.	Maintains accurate faculty publication records for research tracking.	Supports publication management.

FR-003	Functional	The system shall track publication metrics including indexing status and citation information.	Medium	Pass: Indexing status and citation information are displayed correctly for each publication. Fail: Publication metrics cannot be retrieved or are incorrect.	Helps the research dean and faculty monitor publication impact and research performance.	Supports publication metrics tracking.
FR-004	Functional	The system shall allow faculty researchers to submit and manage co-author approval requests for publications.	Medium	Pass: Co-author approval requests can be submitted, reviewed, and their status is recorded. Fail: Approval status cannot be updated or retrieved.	Supports the co-author approval workflow for research publications.	Supports collaborative publication management.
FR-005	Functional	The system shall provide fund burn-up analytics to help the Research Dean monitor grant expenditure and remaining funds.	Medium	Pass: Burn-up analytics display total allocation, expenditure, and remaining funds correctly. Fail: The displayed financial values	Helps the Research Dean monitor research fund utilization and identify potential overspending.	Supports financial monitoring and decision-making.

				are missing or inaccurate.		
NFR-001	Nonfunctional	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals and journal status modifications.	High	Pass: Audit records cannot be modified or deleted without detection and all financial approvals are logged. Fail: An audit record can be altered without detection.	Ensures financial approvals and journal modifications remain secure and traceable.	Security and auditability requirement.
NFR-002	Nonfunctional	The system shall provide publication and grant information within 3 seconds under normal operating conditions. **4. Priority**	Medium	Pass: At least 95% of requests return the required information within 3 seconds during normal operation. Fail: More than 5% of requests exceed 3 seconds.	Provides responsive access to research and financial information for faculty and the Research Dean.	Performance requirement.

Faculty Research Grant & Publication Tracker

UML Component Diagram



Architecture Selection: Layered Architecture

Architecture Selection: We chose Layered Architecture for the Faculty Research Grant & Publication Tracker System.

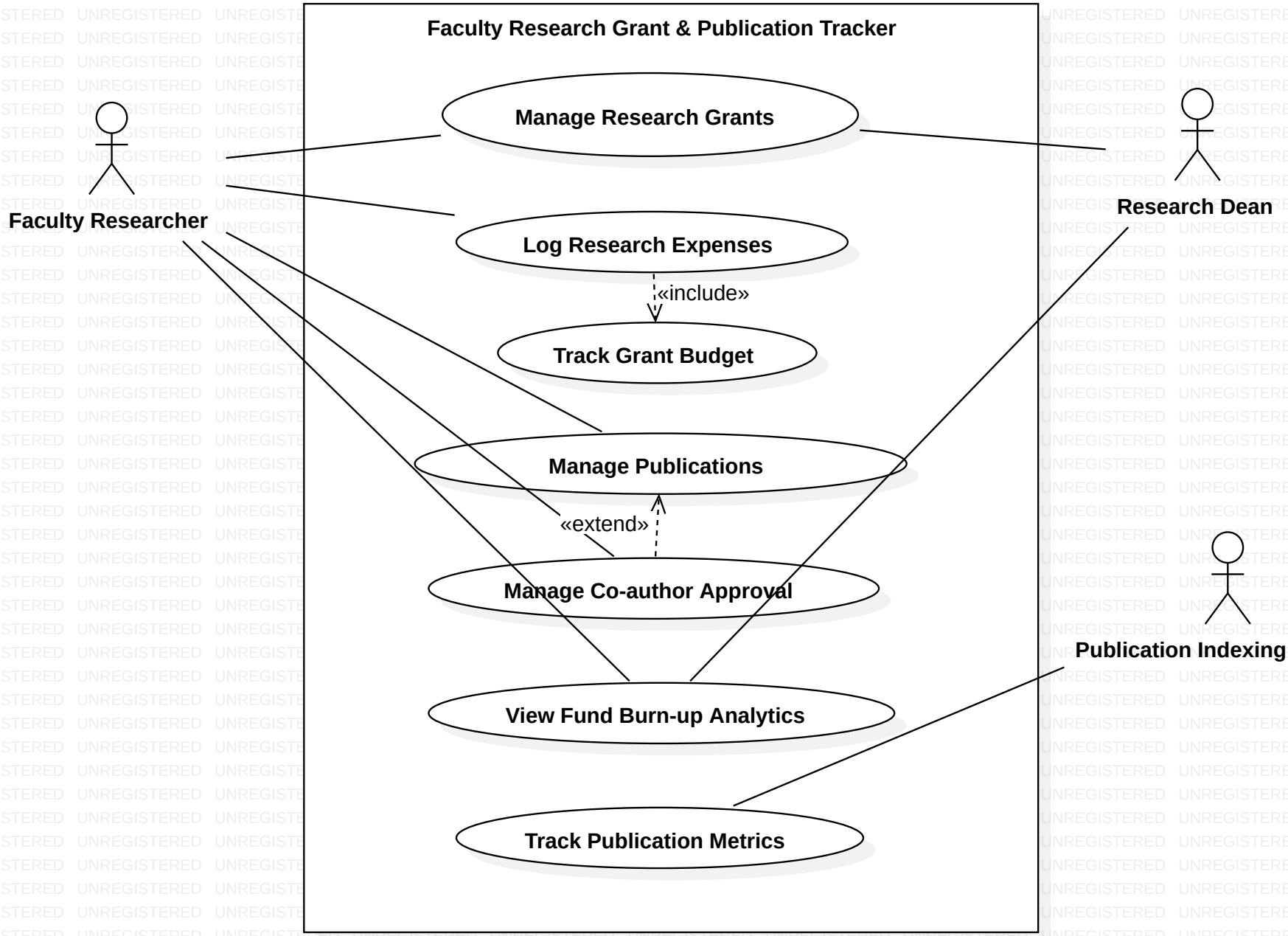
Reason 1: It separates the user interface, business logic and data management, making the system easier to understand and maintain.

Reason 2: It is suitable because the system handles different functions such as grant management, publication management and analytics.

Security Advantage: Security can be improved by keeping database access inside the data layer and controlling access through the business layer.

Performance Benefit: Performance can be improved by keeping frequently used data and operations organized within separate layers.

Model1::UseCaseDiagram1



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Use-Case Flow Specification

Use Case: Log Research Expenses

Primary Actor: Faculty Researcher

Supporting Use Case: Track Grant Budget

Preconditions

1. The Faculty Researcher is authorized to access the research grant system.

2. A valid research grant exists.

3. The grant has an available budget.

Postconditions

1. The research expense is recorded successfully.

2. The remaining grant balance is updated.

3. The expense is included in the grant's financial records.

Main Success Scenario

1. Faculty Researcher selects "Log Research Expenses".

2. The system displays the expense entry form.

3. Faculty Researcher enters the expense details, including amount, category, date, and description.

4. The system validates the entered information.

5. The system checks the available grant balance.

6. The system records the expense.

7. The system calculates the remaining grant budget.

8. The system displays a confirmation message and updated balance.

9. The use case ends successfully.

Alternate Flow – Expense Exceeds Available Grant Balance

1. The system detects that the expense exceeds the available grant balance.

2. The system rejects the expense.

3. The system displays an appropriate error message.

4. The Faculty Researcher can modify the expense amount or cancel the operation.

5. No invalid expense is recorded.